UNIT 4

FINANCIAL ARITHMETIC

REVISION EXERCISE

1. Find the fourth proportional of the following:

   (i) 3,4,6 (ii) 7, 11,14 (iii) 38, 19, 18 (iv) 5.6, 6.3, 4.8

2. 11 and 27 are the extremes of a proportion. If one of the means is 9, find the other?

3. The cost of 25 articles is Rs 300. What will be the cost of 225 such articles?

4. How many metres of cloth will be purchased for Rs 1,315, if the cost of 40 metres of cloth is Rs 1,000?

5. A ship covered the distance of 800 knots in 40 days. In how many days will it cover the distance of 2,440 knots, if its speed remains the same?

6. A watch after every 12 hours goes fast by three minutes. How many minutes, will it go fast in 14 days?

7. The height of a tower is 20 metres and the length of its shadow is 25 metres. What is the height of another tower, whose shadow is 75 metres at the same time?

8. 120 men can do a work in 90 days. How many more men are required to complete the work in 40 days?

9. In a fort 600 persons have food provisions for 27 days and after 10 days 80 more persons joined them. For how many days will the same food provision be sufficient.

10. 200 persons have food provision for 20 days at a camp. How many persons should leave the camp so that the same food provision be sufficient for 25 days?

4.1 Compound Proportion

4.1.1 Define Compound Proportion

The relationship between two or more proportions is known as compound proportion.

There are three cases of compound proportion.

Case 1: Both proportions are direct.

Case 2: One proportion is direct and the other is inverse.

Case 3: Both proportions are inverse.

4.1.2 Solve real life problems involving compound proportion, partnership and inheritance.

A. Solve real life problems involving compound proportion

For more clarification of compound proportion, consider the following examples.

Example 1. If 70 men dig 630 m^3 of soil in 5 days, how much of the soil will 80 men dig in 10 days?

Solution: This example relates to case 1

Increase in men $\Rightarrow$ Increase in soil. ... (Direct Proportion)

Increase in days $\Rightarrow$ Increase in soil. ... (Direct Proportion)

Let x m^3 be the soil dug by 80 men in 10 days.

Men	Days	Soil (m^3)
70 ↑	5 ↑	630 ↑
80	10	x

(Both increase)

$$\Rightarrow \quad \frac{x}{630} = \frac{10}{5} \times \frac{80}{70}$$

$$\Rightarrow x = \frac{\overset{2}{\cancel{10}} \times 80 \times \overset{9}{\cancel{630}}}{\cancel{5} \times \cancel{70}} = 2 \times 80 \times 9 = 1440 \text{ m}^3.$$

Thus, 1440 m^3 soil will be dug out.

Example 2. Rs 12,000 are sufficient for a family of 4 members for 32 days. For how many days will Rs 24,000 be sufficient for a family of 8 members?

Solution: This example relates to case 2.

Increase in Rupees $\Rightarrow$ Increase in Days, i.e., Direct proportion.

Increase in Members $\Rightarrow$ Decrease in Days i.e., Inverse proportion.

Rupees	Members	Days
12,000 ↑	4 ↓	32 ↑
24,000	8	x

Let x be the days for which Rs 24,000 will be sufficient for a family of 8 members.

$$\Rightarrow \quad \frac{x}{32} = \left(\frac{24000}{12000}\right) \times \frac{4}{8} = \frac{\cancel{24000}^{2}}{\cancel{12000}_{1}} \times \frac{\cancel{4}^{1}}{\cancel{8}_{\cancel{4}_1}} = \frac{1}{1}$$

$$\text{or} \quad \frac{x}{32} = \frac{1}{1} \quad \Rightarrow x = 32 \text{ days.}$$

Thus, the given amount of Rs 24000 shall be sufficient for 32 days.

Example 3. If 2,400 men have sufficient food for 20 days at the rate of 1.2 kg per person. How many men should leave so that the same food be sufficient for 40 days at the rate of 1.5 kg per person?

Solution: This example is related to case 3.

Increase in Food per person ⇒ Decrease in Men (Inverse Proportion).

Increase in Days ⇒ Decrease in Men (Inverse Proportion).

Let x be the number of men who have sufficient food for 40 days by taking 1.5 kg per person.

Food per person	Days	Men
1.2 kg ↓	20 ↓	2400 ↑
1.5 kg	40	x

$$\Rightarrow \quad \frac{x}{2400} = \frac{\cancel{20}}{\cancel{40}_{\cancel{2}}} \times \frac{\cancel{1.2}^{\cancel{4}^{2}}}{\cancel{1.5}_{\cancel{5}}} = \frac{2}{5} \qquad \text{(Both decrease)}$$

$$\Rightarrow x = \frac{2}{\cancel{5}_1} \times \cancel{2400}^{480} = 960 \text{ men}$$

Thus, the food will be sufficient for 960 men.

So, 2400 – 960 = 1440 men will leave.

EXERCISE 4.1

1. The fare of 12 kg of luggage for a distance of 36km is 24 rupees. How much fare will be charged for a luggage of 18kg for a distance of 12km?

2. 4800 men have sufficient food for 20 days at the rate of 1.5kg per person. Find the number of men for which the same food will be sufficient for 16 days at the rate of 1.0 kg per person.

3. The price of a carpet 16 metres long and 6 metres wide is Rs 6,288. What will be the price of 24 metres long and 12 metres wide carpet?

4. 10 men can assemble 200 cycles in 8 days. How many cycles, 5 men will assemble in 16 days?

5. If 60 persons use 40kg of sugar in 12 days, find in how many days 90 persons will use 200kg of sugar.

6. 14 persons working 5 hours daily complete a work in 8 days. Find for how many hours per day, 35 persons may work to complete the same work in 4 days?

7. A contractor engaged 30 persons to construct the roads in 30 days, but only one fourth of the work was completed after 10 days. How many more persons he requires to complete the work in time?

8. If fare for carrying out 1200 kilograms of weight at a distance of 200 kilometers is Rs 300, find how much weight can be taken to a distance of 300 kilometers for Rs 500?

9. 3 masons complete flooring 100 meters long and 40 meters broad in 16 days. In how many days will 4 masons complete the flooring 160 meters long and 50 meters broad?

10. 14 persons working 5 hours daily can complete one third of the work in 8 days. How many hours per day are needed to complete the rest of work in 4 days by 35 persons?

B. Partnership: It is a business which two or more persons run it together by sharing the profit and loss.

(i) Simple Partnership: It is the partnership in which, the partners start the business and close it together with equal or different investment capitals.

(ii) Compound Partnership: It is the partnership in which, the partners contribute different capitals for different time periods.

The profit or loss is divided on the basis of the amount of investment and the period of investment of each partner.

Solve real life problems involving partnership.

Example 1. Asad and Ammar started a business with capitals of Rs 63,000 and Rs 72,000, respectively. After one year they earned a profit of Rs 45,000. Find the share of each one in the profit.

Solution: This is the case of simple partnership. As the time period of investment of both partners is the same, so the profit will be divided on the basis of ratio of the their capital share.

Ratio between the investments in rupees:

Asad		Ammar
63,000	:	72,000
63	:	72
7	:	8

Sum of ratios = 7 + 8 = 15

Total profit = Rs. 45,000

$$\text{Asad's profit} = \frac{7}{\cancel{15}} \times \overset{3,000}{\cancel{45,000}} = \text{Rs } 21,000.$$

$$\text{Ammar's profit} = \frac{8^{1}}{\underset{1}{\cancel{15}}} \times \overset{3,000}{\cancel{45,000}} = \text{Rs } 24,000.$$

Example 2. Amjad started a business with a capital of Rs 30,000. After 6 months Umar joined him with an investment of Rs 60,000. Usman joined them after three months by investing Rs 180,000. At the end of the year they earned a profit of Rs 490,000. Find the share of each partner in the profit.

Solution: This is the case of compound partnership in which capitals and time periods of investments of partners are different.

Amjad's effective investment = Investment capital × Time period

= 30, 000 × 12 = Rs 360,000

Umar's effective investment = 60,000 × 6 = Rs 360,000

Usman's effective investment = 180,000 × 3 = Rs 540,000

Ratio of effective capitals of partners in rupees:

Amjad	:	Umar	:	Usman
360,000	:	360,000	:	540,000
36	:	36	:	54
2	:	2	:	3 (Sum of ratios = 2+2+3= 7)

Amjad's Share in profit = $\frac{2}{7} \times 490{,}000$ = Rs 140,000

Umar's Share in profit = $\frac{2}{7} \times 490{,}000$ = Rs 140,000

Usman's Share in profit = $\frac{3}{7} \times 490{,}000$ = Rs 210,000.

(C) Solve real life problems involving Inheritance

When a person dies, the assets left by him or her is called inheritance and it is distributed among his or her legal inheritors(heirs) according to the law of Shariah.

The steps given below will be followed:

- First of all his/her debts will be paid.
- Then as per will upto $\frac{1}{3}$ rd of his/her property will be executed, if asked for in the will.
- Then the remaining inheritance will be distributed among the inheritors as follows:

(i) If parents of the deceased person are alive, each of them will get $\frac{1}{6}$ th part of his/her property.

(ii) Widow (s) will get $\frac{1}{8}$ th part, whereas husband will get $\frac{1}{4}$ th part of the remaining property, if they have children. In case of no child, widow will get $\frac{1}{4}$ th part, whereas husband will get $\frac{1}{2}$ nd part of the remaining property (Brothers and sisters will get the rest).

(iii) Each daughter will get half of the share of each son.

Example 3. Anwer left a property of Rs 1,000,000 in cash and a plot worth Rs 600,000. If he left a widow, two sons and three daughters, how much amount will each receive?

Solution: Total property = cash + a plot

= Rs 1,000,000 + Rs 600,000 = Rs 1,600,000

Widow will receive $= \frac{1}{8} \times 1{,}600{,}000$ = Rs 200,000

Amount left = Rs 1,600,000 – Rs 200,000 = Rs 1,400,000

Let share of a daughter be = 1, then the share of a son will be = 2

Shares of **3** daughters = 3 × 1 = 3 and Shares of **2** sons = 2 × 2 = 4

Sum of shares of children = 4 + 3 = 7.

Each son will receive $= \frac{2}{7} \times 1{,}400{,}000$ = Rs 400,000

Each daughter will receive $= \frac{1}{7} \times 1{,}400{,}000$ = Rs 200,000

Example 4. Mrs. Fatima died leaving behind her a property of Rs 645,000. A debt of Rs 40,000 was due to her and Rs 5,000 was spent on her burial. Distribute the amount among her mother, husband, one daughter and three sons.

Solution: The amount to be distributed among heirs

= Total amount left – Debt Amount – Burial expenses

= Rs 645,000 – Rs 40,000 – Rs 5,000 = Rs 600,000

Mother's will receive $= \frac{1}{6} \times 600{,}000$ = Rs 100,000

Husband's will receive $= \frac{1}{4} \times 600{,}000$ = Rs 150,000

Remaining inheritance = Rs 600,000 – Rs 100,000 – Rs 150,000

= Rs 600,000 – Rs 250,000 = Rs 350,000

Share of a daughter = 1 and share of a son = 2

Shares of 3 sons = $3 \times 2 = 6$ and sum of shares = $1 + 6 = 7$

Daughter will receive = $\frac{1}{7} \times 350{,}000$ = Rs 50,000

Each son will receive = $\frac{2}{7} \times 350{,}000$ = Rs 100,000

EXERCISE 4.2

1. Marium and Arifa started a business with investments of Rs 60,000 and Rs 80,000 respectively. After one year they earned a profit of Rs 21,000. Find the share of each partner in the profit.

2. Akram started a business with Rs 70,000. After 3 months Aslam joined the business with Rs 40,000 and after 6 months Asghar also invested Rs 40,000. At the end of the year they earned a profit of Rs 28,800. Find the share of each partner in the profit.

3. Ali, Zain and Saad started a business with Rs 30,000, Rs 30,000 and Rs 24,000, respectively. After 5 months period Zain withdraws Rs 18,000 and the business is closed after 10 months. How much did each partner receive in the profit of Rs 37,500?

4. Ali and Usman invested different amounts for different periods in a business. If the total profit in a business after one year is Rs 60,000, fill in the following blanks:

S. No.	Investment of Ali (Rs)	Investment of Usman (Rs)	Period of Ali (Months)	Period of Usman (Months)	Ratio of Profits	Profit of Ali (Rs)	Profit of Usman (Rs)
(i)	50,000	25,000	12	12	2:1	40,000	20,000
(ii)	50,000	75,000	12	12	______	______	______
(iii)	25,000	25,000	6	12	______	______	______
(iv)	50,000	25,000	12	6	______	______	______
(v)	50,000	______	6	6	2:3	______	______

5. A man left Rs 1,800,000 as inheritance. His heirs are 6 daughters and 2 sons. Find the share of each heir.

6. Aslam died leaving a property of Rs 1,660,000. He left two widows, two sons and one daughter. Find the share of each heir if the burial expenses were Rs 60,000.

7. Akram left a wealth of Rs 1,520,000. Among his heirs is a mother, a widow, three sons and four daughters. The funeral expenses are Rs 30,000 and a loan of Rs 50,000 is due to him. Calculate the share of each heir.

8. Mrs. Abdul Rehman died leaving behind her father, her mother, two sons and husband. Find the share of each son if the amount of her inheritance is Rs 480,000.

9. A person died leaving behind a number of children (No parents, no wife). If the amount of inheritance is Rs 800,000. Fill in the blanks for different number of sons and daughters.

S. No.	Number of Sons	Number of Daughters	Total Shares of Sons	Total Shares of Daughters	Amount for each Son (Rs)	Amount for each Daughter(Rs)
(i)	1	2	2	2	400,000	200,000
(ii)	2	1	4	1	______	______
(iii)	2	4	______	______	______	______
(iv)	______	6	2	______	______	______
(v)	3	______	______	4	______	______
(vi)	______	______	2	______	______	100,000

4.2 BANKING

The main function of a bank is to receive deposits from many of its depositors and to give loans to its account holders. In return, bank pays small amounts of profits on deposits and charges markup on loans.

4.2.1 Types Of A Bank Account

(a) Define Commercial Bank Deposits

The deposits received from customers and kept in different accounts such as current deposit account, PLS account, foreign currency account etc. are called **commercial bank deposits**.

Following are the types of a Bank Account.

(i) PLS Saving Bank Account

This type of account encourages the account holders to deposit more amount by offering reasonable rates of profit on deposits. The term of the deposit is not fixed. Zakat is deducted on first of holy month RAMZAN.

(ii) Current Deposit Account

It is opened for managing daily transactions. Account holders are required to maintain a minimum amount of money in their accounts. Banks do not offer any profit on current accounts.

(iii) PLS Fixed Deposit Account

In this type of account, money is kept in a bank for a fixed period of time. Banks offer higher profit rates for fixed deposit accounts as compared to non-fixed deposit accounts.

(iv) Foreign Currency Account

Multinational companies or Pakistani nationals living abroad operate such type of account. In this type of account transaction in foreign currency is allowed. Zakat and tax are not deducted on this account.

(b) Describe Negotiable Instruments

It is a written promise to pay a fixed sum of money on demand or at a certain time. Common examples are cheques, pay orders and demand drafts.

(i) Cheque: It is a method of withdrawing amounts from a bank account by account holders. It is a written document addressed to a specified bank.

(ii) Demand Draft: It is an unconditional written order drawn by one bank to another bank, to pay a specified sum of money to a third party on demand.

(iii) Pay order: A negotiable instrument issued by a bank that instructs a bank to pay a certain amount of money to a third party, is called a pay order. Issuing bank is bound to pay the amount.

4.2.2 On-line Banking

(i) Explain On-line banking

On-line banking allows the customers to perform banking transactions, pay utility bills and transfer money from their account at any time, using computers and internet connection.

(ii) Transaction Through Automated Teller Machine (ATM)

Customers are provided ATM card to use as an ATM facility. The ATM software performs transactions when a customer inserts the ATM card in the machine and provides some details which include the PIN (Personal Identification Number).

(iii) Debit Card: These are provided by the bank and are used to buy goods or withdraw cash. The amount is taken from the account holder's account.

(iv) Credit Card: These are plastic cards with a magnetic strip storing information about the account holder which allows purchases locally.

4.2.3 Conversion of Currencies

Exchange rates are used to convert currencies. These rates show the relationship between the value of different currencies and are not permanent.

(a) Convert Pakistani Currency to well-known International Currencies

Foreign currency exchange rate is a price that determines how much it costs to buy the currency of a foreign country in Pakistani currency.

Table below shows some *current exchange rates of the currencies.

Country	Currency	Symbol	Buying (PKR)	Selling (PKR)
US	Dollar	$	103.40	103.65
UK	Pound	£	139.00	139.75
Saudi Arabia	Riyal	ریال	27.15	27.40
India	Rupee	₹	1.60	1.65
European (Common Market)	Euro	€	126.00	126.25
China	Yuan	¥	16.25	16.50
Japan	Yen	¥	0.946	0.953

** These exchange rates were valid at the time of writing this book (28.11.2016). These may vary from time to time and can be seen in electronic and/or print media.*

Example 1. An American wants Pakistani rupees in exchange of 250 dollars. How many rupees will he get? **(1 US dollar ($) = 103.40 PKR).**

Solution: 1 US dollar ($) = Rs 103.40

250 US dollars ($) = 250×103.40 = Rs 25,850.

Example 2. Mr. Jamal wants Saudi Riyals in exchange of 60,000 Pakistani rupees. How many Saudi Riyals will he receive?

(1 Saudi Riyal ریال = 27.15 PKR).

Solution: Amount to be converted = Rs 60,000

Rate of conversion is 1 Saudi Riyal ریال = Rs 27.15 (Pakistani)

The number of Saudi Riyals received = $\dfrac{60,000}{27.15}$ = 2,209

EXERCISE 4.3

1. Convert:

(i) Rs 90,000 into Chinese Yuan (1 Yuan = Rs 16.25)

(ii) Pakistani Rs 50,000 into Indian Rupees

(1 Indian Rupee = Rs 1.60 Pakistani)

(iii) 110 UK Pounds into Pakistani Rupees **(1 UK Pound(£)= Rs 139 Pakistani).**

(iv) 250 Euro into Pakistani rupees **(1 Euro(€)= Rs 126.0 PKR).**

(v) Rs 64,200 Pakistani into Turkish Lira (1 Turkish Lira = Rs 42.80 Pakistani).

2. A Pakistani serving in Saudi Arabia earns 9,000 Riyals a month. He spends 2500 Riyals a month. Determine his monthly saving in Pakistani rupees, if 1 Riyal = Rs 27.15 Pakistani.

3. Kamal has Rs 38,400 (Pakistani) and 4,000 Chinese Yuans and he wants to exchange them for U.S Dollars. How many dollars will he get if the exchange rate is: 1 U.S Dollar = Rs 103.40 Pakistani and 1 Chinese Yuan(¥) = Rs 16.25 Pakistani?

4. Convert Pakistani Rs 48,000 into Indian Rupee.

5. Convert 48,000 Indian rupees into Pakistani rupees.

6. Price of a fridge in a Pakistani market is Rs 77,500. The same fridge in Japan is being sold for 70,000 Yen. Transporting expenses are Rs 5,000 and custom duty is Rs 3,000. How much will it cost if it is imported from Japan and how much money would be saved if exchange rate is Rs 0.946 = 1 Yen?

4.2.4 Profit/Markup

(i) **a. Profit (I):** When we deposit money into a bank, the bank uses our money and in return after some time pays an extra amount along with our actual deposit. This extra money which the bank gives us is called profit on the deposit.

(i) **b. Markup (I):** When we borrow money from a bank, the bank charges some extra amount along with the actual money borrowed. This extra money which the bank charges is called markup.

(ii) **Principal amount (P):** It is the amount which we borrow from or deposit in the bank or the amount used.

(iii) **Profit/Markup Rate (R):** It is the rate at which the bank gives profit on deposits or collects markup from its account holder borrowers. This rate is calculated as a percentage of the principal amount. Deposited or borrowed from the bank.

(iv) **Period (T):** It is the time period for which an amount is borrowed or invested.

4.2.4 (v) Calculate the profit/Markup, the principal amount, the profit or markup rate and the time period.

(A) Calculate Profit/Markup (I)

For calculation of profit or markup, we use the formula: **I = PRT**

i.e., Profit or Markup = Principal Amount × Rate × Time Period

Example 1. Akbar borrowed Rs 70,000 from a bank at the rate of 7% for 3 years. Find the amount of markup and the total amount to be paid by Akbar.

Solution: Here Principal Amount (P) = Rs 70,000,

Rate (R) = 7% Time (T) = 3 years

$\therefore$ Markup (I) = $P \times R \times T = 70{,}000 \times \frac{7}{100} \times 3 = 700 \times 7 \times 3 =$ Rs 14,700

Hence, Total Amount to be paid = Amount Borrowed + Markup

= Rs 70,000 + Rs14,700 = Rs 84,700

(B) Calculate Principal Amount (P)

Using $\mathbf{I = PRT}$, we get $\mathbf{P = \frac{I}{R \times T}}$

Example 2. Find the amount invested by Riaz in a business, if he receives a profit of Rs 27,000 at the rate of 12% per annum for 3 years.

Solution: Here I = Rs 27,000, R = 12%, T = 3 years

$$\therefore \quad \mathbf{P} = \frac{\mathbf{I}}{\mathbf{R \times T}} = \frac{\overset{\overset{3000}{\cancel{9000}}}{\cancel{27{,}000}} \times \overset{25}{\cancel{100}}}{\underset{\cancel{3}_1}{\cancel{12}} \times \cancel{3}_1} = \text{Rs } 75{,}000$$

Thus, the amount invested by Riaz = Rs 75,000

(C) Calculate Profit or Markup Rate

Using I = PRT, we get $\mathbf{R = \frac{I}{P \times T}}$

Example 3. At what annual rate percent of markup a principal amount of Rs 34,000 would become Rs 43,180 in 3 years?

Solution: Markup (I) = Total Amount – Principal Amount

= Rs 43,180 – Rs 34,000 = Rs 9,180

As, time T = 3 years; so $R = \frac{I}{P \times T} = \frac{9180}{34000 \times 3} = 0.09$

Hence, the Markup rate in percentage form = $0.09 \times 100 = 9\%$

(D) To Calculate the Time Period (Using $I = PRT$, we get $T = \frac{I}{P \times R}$)

Example 4. In what time would a sum of money become triple of itself at the rate of 5% markup?

Solution: Suppose the principal amount = P = Rs 100.

Triple of Amount = 3P = Rs 300.

Markup (I) = Total Amount − Principal Amount

$= 3P - P = 2P \quad = 300 - 100 = \text{Rs. } 200$

Here, $R = 5\% = \frac{5}{100} = 0.05.$

$$\therefore \quad T = \frac{I}{P \times R} = \frac{200}{100 \times 0.05} = \frac{200}{5} = 40 \text{ years.}$$

Example 5. How long would Rs 25,000 have to be deposited in a bank at 12% markup per annum to receive back Rs 34,000?

Solution: Principal Amount = P = Rs 25,000, Amount Received = Rs 34,000

Markup (I) = Total Amount − Principal Amount

$= 34{,}000 - 25{,}000 = \text{Rs } 9{,}000$

As, $R = 12\% = \frac{12}{100}$, So $T = \frac{I}{P \times R} = \frac{\overset{3}{\cancel{9000}} \times \overset{1}{\cancel{\overset{4}{\cancel{100}}}}}{\underset{1}{\cancel{25000}} \times \underset{1}{\cancel{\underset{3}{\cancel{12}}}}} = 3$ years.

4.2.5 Types of Finance

Explain

(i) Over Draft (OD): It is a borrowing facility provided by the bank to account holders, businessmen, companies etc., to withdraw some amount in excess of his/her original account balance, only once.

(ii) Running Finance (RF): It is similar to over draft. It can be considered as a credit facility provided for a certain limit with a variable profit rate. It can be used again and again.

(iii) Demand Finance (DF): It is a type of loan that may be called in by the bank (or lender) at any time. It may be either for a short time or a long time.

(iv) Leasing: Lease is an agreement between the lessee (user) to pay the leaser (owner) for the use of an asset. The ownership of the leased asset during the leased period, known as term, remains with the leaser. The asset will be returned back to the lessee after agreed time.

Hire purchase is a method of buying goods in which payments of purchased price is spread over specific term by payment of an initial deposit called **down payment**. Goods will not be returned back.

4.2.5 Solve Real Life Problems Related To Banking And Finance.

Example 1. A company gets a house on lease for 6 years. According to an agreement the company paid Rs 2,000,000 as down payment and shall pay Rs 40,000 per month as rent per month. After 3 years the company shall increase the rent by 5%. Calculate total amount the leaser (owner) would get.

Solution: Down payment received by the owner = Rs 2,000,000

Rent per month for the first 3 years = Rs 40,000

$\therefore$ Rent for the first 3 years $= 3 \times 12 \times 40{,}000 =$ Rs 1,440,00

Rent per month after 3 years $= 40{,}000 \times \dfrac{105}{100} =$ Rs 42,000

$\therefore$ Rent for next three years $= 3 \times 12 \times 42{,}000 =$ Rs 1,512,000

Hence, total amount received by the owner

= Down payment + rent for first 3 years + Rent for next 3 years

$= 2{,}000{,}000 + 1{,}440{,}000 + 1{,}512{,}000 =$ Rs 4,952,000

Example 2. The price of a car is Rs 900,000. It can be bought at 15% of the price as down payment. It had to be leased on a simple markup of 10% per year for 2 years on the remaining amount. The instalments will be made on monthly basis.

Find: (i) The monthly instalment and

(ii) The total leased price of the car to be paid.

Solution: Down payment = 15% of Rs 900,000 $= \dfrac{15}{\cancel{100}} \times 900{,}\cancel{000}$

$= 15 \times 9000 =$ Rs 135,000

The remaining amount (P)= Rs 900,000 – Rs 135,000 = Rs 765,000

$\therefore$ Markup for 2 years on the remaining amount is given by:

$I = P \times R \times T \quad = 76500\cancel{0} \times \frac{1\cancel{0}}{1\cancel{00}} \times 2 = \text{Rs } 153,000$

So, the amount to be paid in 24 monthly instalments

= Remaining Amount + Markup (I) = Rs 765,000 + Rs 153000

= Rs 918,000

$\therefore$ Monthly instalment = Rs 918,000 ÷ 24 = Rs 38,250

Total Amount Paid = Down Payment+ Remaining Amount+Markup

=Rs 135,000 + Rs 765,000 + Rs 153,000 = Rs 1,053,000

EXERCISE 4.4

1. Find the profit on Rs 50,000 at the rate of 5% per year for 6 years.

2. Asad borrowed Rs 25,000 from a bank at the rate of 8% per annum for 4 years. Find the markup of the bank.

3. Naseem receives a profit of Rs 27,000 at the rate of 12% per year for 3 years. Find his original investment.

4. At what annual rate percent would Rs 68,000 amount to Rs 90,440 in 11 years?

5. How long would Rs 31,000 have to be invested at a markup rate of 6% per year to gain Rs 5,332?

6. At what rate of markup would a sum of money become double in 20 years?

7. In what time would a sum of money will double itself at 8% of markup?

8. Complete the following table:

S. No.	Markup **(I)** (Rupees)	Principal Amount **(P)** Rupees	Time **(T)** (Years)	Rate of Markup **(R)** (%)
(i)	________	20,000	5	4
(ii)	________	98,000	6	6.5
(iii)	10,500	21,000	________	5
(iv)	2,100	________	6	7
(v)	1,740	5,800	3	________

9. Hashim buys an air-conditioner for Rs 50,000. To lease it, he has to pay 20% down payment and the remaining amount on a simple markup of 10% per year for 2½ years on monthly instalments.

Find: (i) Monthly instalment (ii) Total amount paid.

10. A company gets a house on lease for 4 years. The company paid Rs 800,000 as down payment and agreed to pay Rs 20,000 per month as rent. After 2 years the company agreed to increase the rent by 5%. Calculate the total amount that the owner (leaser) would get.

11. Arshad buys a flat for Rs 200,000 on lease. Complete the following table for different mark up rate, time period etc.

S. No.	Down Payment (Percent)	Markup rate per year (Percent)	Period of Lease (in Years)	Monthly Installment (in Rupees)	Total amount paid (in Rupees)
(i)	20%	10%	2	______	______
(ii)	20%	5%	2	______	______
(iii)	20%	10%	4	______	______
(iv)	40%	10%	2	______	______

4.3 PERCENTAGE

The percentage means "per hundred" or "out of hundred". The symbol for percentage is %. For example 40% means 40 out of 100.

4.3.1 Profit and Loss

If the selling price (S.P.) is higher than the cost price (C.P.), then the profit occurs. Profit is the difference between S.P. and C.P., i.e.,

Profit = S.P. – C.P.

If the cost price (C.P.) is higher than the selling price (S.P.), then loss occurs. Loss is the difference between C.P. and S.P., i.e.,

Loss = C.P. – S.P.

4.3.1 (i) Find percentage profit and percentage loss

Profit and loss is always on the cost price.

Hence, percentage profit or loss is expressed in terms of cost price. Thus,

Profit percentage = $\frac{\text{Profit}}{\text{C.P.}} \times 100$. Loss percentage = $\frac{\text{Loss}}{\text{C.P.}} \times 100$.

Example 1. If C.P = Rs 1,000 and Profit = Rs 150, then

Profit percentage = $\frac{\text{Profit}}{\text{C.P.}} \times 100 = \frac{150}{1000} \times 100 = 15\%$

Example 2. If C.P = Rs 2,000 and loss = Rs 100, then

Loss percentage = $\frac{\text{Loss}}{\text{C.P.}} \times 100 = \frac{100}{2000} \times 100 = 5\%$

4.3.2 Discount

In order to attract customers, some times price of an article is reduced and is sold at a price lower than its marked or listed price. The difference between marked price (M.P.) and selling price (S.P.) of an article is called discount. Thus,

Discount = Marked price – Sale price = M.P. – S.P.

Discount % = $\frac{\text{Discount}}{\text{M.P.}} \times 100$

Discount percentage is also known as **rate of discount**.

(ii) Find Percentage Discount

Example 1. Marked Price (M.P) = Rs 3,000, Sale Price (S.P) = Rs 2,700
Therefore, Discount M.P – S.P = Rs 3,000 – 2,700 = Rs 300.

Discount % = $\frac{\text{Discount}}{\text{Marked Price}} \times 100 = \frac{300}{3000} \times 100 = 10\%$.

Example 2. S.P = Rs 1,500, Discount = Rs 200

$\therefore$ M.P = S.P+ Discount = Rs 1,500+Rs 200= Rs 1700

Discount % = $\frac{\text{Discount}}{\text{Marked Price}} \times 100 = \frac{200}{1700} \times 100 = 11.76\%$.

(iii) Solve problems involving Successive Transactions (Chain Discount)

If the discounts are deducted from the marked price one after the other, it is known as chain discount (like successive transactions).

Note: Rates of discount may be different or same each time.

In chain discount, we find the amount of the first discount on the marked price and find the first S.P. Then we find the amount of second discount on the first S.P. to obtain the second S.P. and so on.

Example 1. Find the net selling price of a sofa set listed at Rs 80,000, if chain discounts of 8% and 4% are allowed.

Solution: List price (M.P.) = Rs 80,000

First discount = 8% of Rs 80,000 = $\dfrac{8}{100} \times 80{,}000 = \text{Rs } 6{,}400$

$\therefore$ First S.P. = List price – First discount = 80,000 – 6,400 = Rs 73,600

Second discount = 4% of First S.P. = $\dfrac{4}{100} \times 73{,}600 = \text{Rs } 2{,}944.$

Therefore, the second or net final S.P. = First S.P – Second discount

= 73,600 – 2,944 = Rs 70,656

Note: Students are advised to check that final S.P of the sofa will be the same, if successive discount rates are 4% and 8% (i.e., Order of discount rates does not matter).

Example 2. Asad bought a car for Rs 500,000 and sold it for Rs 550,000. Find his percentage profit.

Solution: C.P. = Rs 500,000, S.P. = Rs 550,000

$\therefore$ Profit = S.P. – C.P. = 550,000 – 500,000 = Rs 50,000

and profit % = $\dfrac{\text{Profit}}{\text{C.P.}} \times 100 = \dfrac{50{,}000}{500{,}000} \times 100 = 10\%$

Example 2. Jamal bought a house for Rs 333,000 and sold it for Rs 283,050. Find his loss percentage.

Solution: C.P = Rs 333,000, S.P. = Rs 283,050

$\therefore$ Loss = C.P. – S.P. = 333,000 – 283,050 = Rs 49,950

Hence, loss % = $\dfrac{\text{Loss}}{\text{C.P}} \times 100 = \dfrac{\overset{\overset{15}{\cancel{555}}}{\overset{1665}{\cancel{49{,}950}}}}{\underset{\underset{1}{\cancel{37}}}{\underset{\cancel{111}}{\cancel{333{,}000}}}} \times 1\cancel{00} = 15\%$

Example 3. The marked price of an article is Rs 1,800. After discount, the article is sold at Rs 1,620. Find the percentage discount.

Solution: Marked price = Rs 1,800, Sale price = Rs 1,620

$\therefore$ Discount = M.P. – S.P. = 1,800 – 1,620 = Rs 180

Hence, Discount % $= \dfrac{\text{Discount}}{\text{Marked Price}} \times 100 = \dfrac{180}{1800} 100 = 10\%$

Example 4. Kamal bought some article whose marked price was Rs 5,000. He was allowed 20% discount on his purchase. Find the sale price of that article.

Solution: Marked price = Rs 5,000, Rate of Discount = 20%

$\therefore$ Discount $= 5{,}000 \times \dfrac{20}{100}$ = Rs. 1,000

Hence, Sale price = M.P. – Discount = 5,000 – 1,000 = Rs 4,000.

Example 5. The cost price of a toy is Rs 3,000. The shopkeeper writes the marked price (labelled) 15% above the cost price but sales it in Rs 2,760. Find the percentage discount given to the customer.

Solution: C.P. = Rs 3,000, Increase = 15% , S.P = Rs 2,700

Total increase in C.P. $= \dfrac{3{,}000 \times 15}{100}$ = Rs 450

Therefore, Marked price = C.P. + Total increase in C.P

= 3,000 + 450 = Rs 3,450

S.P = Rs 2,760

Discount = M.P. – S.P. = 3,450 – 2,760 = Rs 690

Percentage Discount $= \dfrac{\text{Discount}}{\text{Marked Price}} \times 100$

$= \dfrac{690}{3450} \times 100 = \dfrac{\cancel{690}^{1}}{\cancel{3450}_{5}} \times 100 = \dfrac{100}{5} = 20\,\%$

Example 6. A whole seller sold a motor-cycle to a retailer at a profit of 10%. The retailer sold it for Rs 37,950 at a profit of 15%. What is the cost price of the whole seller?

Solution: First of all we will find the C.P. for the retailer (which is the S.P. for the whole seller).

S.P for the retailer = Rs 37,950   Profit percentage for the retailer = 15%

Let C.P. for the retailer = Rs 100

S.P. for the retailer = C.P. + Profit % = 100 + 15 = Rs 115

If the S.P. for the retailer is Rs 115, his C.P. = Rs 100

If the S.P. for the retailer is Re 1, his C.P. = $\frac{100}{115}$

If the S.P. for the retailer is Rs 37,950, his C.P. will be:

$$\frac{100}{115} \times 37{,}950 = \frac{100}{\cancel{115}_{\cancel{23}\,1}} \times \overset{\overset{330}{\cancel{7590}}}{\cancel{37950}} = \text{Rs } 33{,}000$$

$\therefore$ S.P. for the whole seller = C.P. for the retailer=Rs.33,000

Now we have to find the C.P. for the whole seller.

Let C.P. for whole seller = Rs 100 and % Profit of whole seller = 10%

Then S.P. for the whole seller = 100 + 10 = Rs 110

If S.P. for the whole seller is Rs 110, his C.P. = 100

If S.P. for the whole seller is Re 1, his C.P. = $\frac{100}{110}$

If S.P. for whole seller is Rs. 33,000,

then his C.P. = $\frac{100}{\cancel{110}_1} \times \overset{300}{\cancel{33{,}000}} = \text{Rs } 30{,}000$

$\therefore$ C.P. for the whole seller is Rs 30,000

EXERCISE 4.5

1. Complete the following table:

S. No.	C.P. in Rupees	S.P. in Rupees	Profit in Rupees	% Profit
(i)	1,050	1,155	______	______
(ii)	1,665	______	333	______
(iii)	6,000	______	______	22
(iv)	______	1,250	______	25

2. A shopkeeper purchased 160 chairs at the rate of Rs 900 per chair. He sold 60 chairs at the rate of Rs 1,000 per chair and remaining chairs at the rate of Rs 960 per chair. Find the profit or loss percentage.

3. Jalil purchased 4 old cars for Rs 900,000. He sold these cars for Rs 225,000, Rs 250,000, Rs 300,000 and Rs 215,000, respectively. Calculate the profit or loss percentage of Jalil.

4. Khalil bought a cartoon of 250 eggs for Rs 2,200. Out of these 20 eggs were rotten and 10 broken during the transportation. He sold the remaining eggs for Rs 11 each. Find the profit or loss percentage.

5. Complete the following table when marked price of an article is Rs 10,000 and is sold for different successive discounts.

S. No.	Successive Discount %	Discount (in Rupees)	Sell Price (in Rupees)	Equivalent Simple Discount %
(i)	5% and 15%	500 + 1425	9,075	19.25
(ii)	15% and 5%	________	________	________
(iii)	15% and 10%	________	________	________
(iv)	18% and 12%	________	________	________

6. The price of a bicycle is listed as Rs 8,500. The whole seller allows the retailer chain discounts of 10% and 5%. Find S.P. Of the whole seller.

7. Find a single discount percentage equivalent to successive discounts of 20%, 10% and 5%.

8. A manufacturer sells an article which costs him Rs 5,000 at 20% profit. The purchaser sells the article at 25% gain. Find final sale price of the article.

4.4 INSURANCE

4.4.1 Define Insurance

Insurance is a symbol of protecting or safeguarding against risk or injuries. It is a contract between two parties. A person or a party which agrees to pay an amount on monthly, quarterly or yearly basis to the insurance company is said to be **'insured'**. The insurance company provides financial protection for property, life and health losses or damages.

This contract is called **'insurance policy'**. The instalment to be paid is called **'premium'**. The time period agreed upon by both the parties is called **'maturity'**.

4.4.2 Solve Real Life Problems Regarding Life and Vehicle Insurance

(i) Life Insurance.

Life insurance is an agreement between the insured person and the insurance company for an agreed time period. Policy owner agrees to pay regular instalments of premium to the insurance company which in return agrees to pay back a sum of money at the end of agreed period or on the death or critical illness of the policy owner. The amount of premium and the time of maturity are fixed according to the age of the insurer as per rules of the company.

Example 1. A man purchased a life insurance policy for Rs 400,000. The annual premium is 4.5% of the policy amount, whereas policy fee is at the rate of 0.25%. Calculate the annual premium and quarterly premium at 27% of the annual premium.

Solution: Policy Amount = Rs 400,000

$$\text{Policy fee} = \frac{0.25}{100} \times 400000 = \text{Rs } 1000$$

$$\text{First premium} = \frac{4.5}{100} \times 400000 = \text{Rs } 18{,}000$$

$$\therefore \quad \text{Annual premium} = \text{First Premium} + \text{Policy fee}$$

$$= 18000 + 1000 = \text{Rs } 19{,}000$$

$$\text{Quarterly premium} = \frac{27}{\cancel{100}} \times 19000 = \text{Rs } 5{,}130.$$

(ii) Vehicle Insurance.

Sometimes persons or companies get insurance policies against their vehicles to cover the risk of the theft, accident, fire, etc. In this case premium depends upon factors such as type of the vehicle, cost of the vehicle, age of the driver, etc. The premium is decided by the insurance company at different rates for different periods.

The first premium is usually the total amount of one year instalment.

Example 1. Akber got his car insured for one year, at an insurance rate of 4.5% per annum. If the price of the car is Rs 800,000, find the amount of premium.

Solution: Price of the car = Rs 800,000 and Rate of Insurance = 4.5%

$\therefore$ Amount of premium $= \frac{4.5}{1\cancel{00}} \times 800,\cancel{000} =$ Rs 36,000.

Example 2. Faheem got his bus insured at a rate of 3% per annum for 2 years. The worth of his bus is Rs 4,000,000. Find the total amount paid as premium if the rate of depreciation is 10% per year.

Solution: Worth of the bus = Rs 4,000,000

Rate of annual insurance = 3%, Rate of Depreciation = 10%

$\therefore$ First premium $= \frac{3}{1\cancel{00}} \times 40000\cancel{00} =$ Rs 120,000

Depreciation after one year $= \frac{10}{100} \times 40000\cancel{00} =$ Rs 400,000

Depreciated price after one year = Original price – Depreciation

= Rs 4,000,000 – Rs 400,000 = Rs 3,600,000

$\therefore$ Second premium $= \frac{3}{1\cancel{00}} \times 36000\cancel{00} =$ Rs 108,000

Total amount paid as premium = First premium + Second Premium

= Rs 120,000 + Rs 108,000 = Rs 228,000

EXERCISE 4.6

1. If the amount of life premium is calculated as:

Yearly = 4.75% of the policy amount + policy fee

Policy fee = 0.2% of the policy amount

Half yearly premium = 52% of yearly premium

Quarterly premium = 27% of yearly premium
Monthly premium = 9% of yearly premium,
then complete the table below:

S. No.	Amount of Policy (Rs)	Yearly Premium (Rs) 4.75% + 0.2%	Half Yearly Premium (Rs) (52% of Yearly)	Quarterly Premium (Rs) (27% of Yearly)	Monthly Premium (Rs) (9% of Yearly)
(i)	100,000	______	______	______	______
(ii)	150,000	______	______	______	______
(iii)	300,000	______	______	______	______

2. Hameed got a life insurance policy of Rs 300,000. Find the first premium he has to pay when the rate of annual premium is 5.2% and policy fee is 0.25% .

3. Basheer insured his car for Rs 500,000, at a rate of 2% per annum for 3 years. The depreciation rate is 5% per year. Find the total amount he has to pay as premium.

4. If the amount of vehicle premium calculated yearly is 4.0% of the worth of the vehicle and Depreciation rate = 10% per annum.

Time period = 3 years, then complete the table given below:

S. No.	Worth of Vehicle (Rs in Thousands)	First Premium (Rs in Thousands)	Second Premium (Rs in Thousands)	Third Premium (Rs in Thousands)	Total Premium (Rs in Thousands)
(i)	100	4	3.60	3.24	10.84
(ii)	250	10	______		______
(iii)	300	______	______	______	______
(iv)	______	______	______	16.20	______
(v)	______	______	36	______	______

5. A lady insured her self at a premium rate of 5% for one year. She paid Rs 25,000 as premium. How much is the value of her insurance?

4.5 INCOME TAX

4.5.1 Explain Income Tax, Exempt Income and Taxable Income.

(a) Income Tax: Income tax is imposed by the government on the annual income of a person or company whose income exceeds a certain limit. The rules and rates for income tax are amended by the government and are announced during the presentation of annual budget.

(b) Exempt Income: It is the income which is not subject to income tax under the present income tax law. Exempted income is as follows:

(i) Agriculture Income (ii) Gratuity or pension

(iii) Income of house or property owned by widows.

(c) Taxable Income: It is the difference of total annual income and exempted income.

Taxable Income = Total Annual Income – Exempted Income.

(d) Rebate: Government announces a certain limit of earning on which a person has not to pay income tax. This limit is called rebate. For example, if annual income of a salaried person is Rs 500,000, the rebate is Rs 400,000, then he has to pay income tax only on Rs 100,000, at the rate announced every year in the budget.

(e) Assessment Period: It starts on first July every year and ends on 30th June the following year.

Taxable Income Slabs 2016 - 2017

S. No.	Taxable Income	Rate of Tax
1.	*Where the taxable income does not exceed Rs 400,000*	0%
2	*Where the taxable income exceed Rs 400,000 but does not exceed Rs 500,000*	2% of the amount exceeding Rs. 400,000
3.	*Where the taxable income exceed Rs 500,000 but does not exceed Rs 750,000*	Rs 2,000 + 5% of the amount exceeding Rs 500,000
4.	*Where the taxable income exceed Rs 750,000 but does not exceed Rs 1,400,000*	Rs 14,500 + 10% of the amount exceeding Rs 750,000

S. No.	Taxable Income	Rate of Tax
5.	*Where the taxable income exceed Rs 1,400,000 but does not exceed Rs 1,500,000*	Rs 79,500 + 12.5% of the amount exceeding Rs 1,400,000
6.	*Where the taxable income exceed Rs 1,500,000 but does not exceed Rs 1,800,000*	Rs. 92,000 + 15% of the amount exceeding Rs 1,500,000
7.	*Where the taxable income exceed Rs 1,800,000 but does not exceed Rs 2,500,000*	Rs 137,000 + 17.5% of the amount exceeding Rs 1,800,000
8.	*Where the taxable income exceed Rs 2,500,000 but does not exceed Rs 3,000,000*	Rs 259,500 + 20% of the amount exceeding Rs 2,500,000
9.	*Where the taxable income exceed Rs 3,000,000 but does not exceed Rs 3,500,000*	Rs 359,500 + 22.5% of the amount exceeding Rs 3,000,000
10.	*Where the taxable income exceed Rs 3,500,000 but does not exceed Rs 4,000,000*	Rs 472,000 + 25% of the amount exceeding Rs 3,500,000
11.	*Where the taxable income exceed Rs 4,000,000 but does not exceed Rs 7,000,000*	Rs 597,000 + 27.5% of the amount exceeding Rs 4,000,000
12.	*Where the taxable income exceed Rs 7,000,000*	Rs 1,422,000 + 30% of the amount exceeding Rs 7,000,000

4.5.2 Solve Simple Real Life Problems Related to Individual Income Tax Assesses

Example 1. Annual income of Junaid is Rs 480,000. Calculate the amount of income tax to be paid by Junaid.

Solution: Annual income = Rs 480,000

This income is listed in the Taxable income slab at Sr.# 2, where Rate of Tax is 2% of the amount exceeding Rs 400,000

Here minimum income tax is 0 (Rs 0.0)

Income exceeding Rs 400,000 is:

Rs 480,000 – Rs 400,000 = Rs 80,000

$\therefore$ Income tax = Rate of Tax $\times$ Income exceeding Rs 400,000

$= 2\% \times 80,000 = \frac{2}{100} \times 80,000 = \text{Rs } 1,600$

Example 2. Annual income of Ashraf is Rs 900,000. Calculate the amount of income tax, if he paid Rs 50,000 as Zakat.

Solution: Annual income = Rs 900,000, Amount Paid as Zakat = Rs 50,000

$\therefore$ Income calculated for tax = 900,000 – 50,000 = Rs 850,000

This income is listed in Taxable income slab at Sr. # 4, where rate of tax is 10% of the amount exceeding Rs 750,000 which is

Rs 850,000 – Rs 750,000 = Rs 100,000

Also minimum income tax is Rs 14,500

So, Total Income Tax = Minimum Income Tax + Tax on Rs 100,000

= Rs 14500 + 10% of Rs 100,000

= 14,500 + 10,000 = Rs 24,500.

$\therefore$ Ashraf has to pay Rs 24,500 as income tax.

Example 3. Harmain paid Rs 107,000 as income tax in a year. Find her annual income.

Solution: Income tax paid in a year = Rs 107,000

As this amount is greater than Rs 92,000 by Rs, 15,000 but less a than Rs 137,000, so taxable income of Harmain falls in income slab at Sr # 6, where rate of tax is 15% of the amount exceeding Rs 1,500,000.

Thus we have to find the amount exceeding Rs 1,500,000 on which Harmain paid Rs 15,000 more than Rs 92,000 as income tax.

If Rs 15 is income tax on Rs 100 then Re 1 is income tax on Rs $\frac{100}{15}$

$\therefore$ Rs 15,000 is the income tax on Rs $\frac{100}{15} \times 15{,}000$ = Rs 100,000

Hence, annual income of Harmain is: Rs 1,500,000 + Rs 100,000

= Rs 1,600,000.

Verification: Income tax on Rs 1,600,000 = Rs 92,000 + 15% of 100,000

= Rs 92,000 + 15,000 = Rs 107,000 **(Verified)**

EXERCISE 4.7

1. Monthly salary of Amjad is Rs 75,000. Compute his annual income tax.

2. Hania earns Rs 495,000 in a year. Calculate her income tax, if she paid Rs 40,000 as Zakat.

3. A person has earned Rs 7,000,000 in a year. The Tax deducted at source is Rs 120,000 and Zakat deducted is Rs 130,000. Calculate the income tax which he has to pay at the end of the financial year.

4. The monthly salary of Ismail is Rs 200,000. Find the income tax he paid, if Rs 80,000 and Rs 20,000 are paid as Zakat and wealth tax, respectively.

5. Sufyan paid Rs 497,000 as income tax in a year. Find his monthly income.

6. Bushra paid Rs 289,500 as income tax at the end of a year. If she has paid Rs 130,000 as wealth tax and Rs 120,000 as Zakat, find her annual income.

7. Complete the following table for different income categories:

S. No.	Yearly Income (Rs in 1000)	Income Tax Slab Number	Min. Income Tax (Rs)	Lower Limit (Rs)	Income Exceeding Lower Limit (Rs in 1000)	Tax Rate (%)	Total Income Tax (Rs)
(i)	578	3	2,000	500,000	78	5%	5,900
(ii)	480	______	______	______	______	______	______
(iii)	900	______	14,500	______	______	______	______
(iv)	6,000	11	______	______	2,000	27.5%	______

REVIEW EXERCISE 4

1. Encircle the correct answer:

(i) The concession on a written price is called:

(a) Tax (b) Profit (c) Discount (d) Loss

(ii) The relation between two or more proportions is called:

(a) Direct Proportion (b) Indirect Proportion

(c) Inverse Proportion (d) Compound Proportion

(iii) ATM stands for:

(a) Account Telly Machine (b) Account Transfer Machine

(c) Automated Teller Machine (d) Auto Cash Transfer Machine

(iv) The rate of Zakat is:

(a) 5% (b) 2.5% (c) 10% (d) 1.0%

(v) The cost price of a book is Rs. 300. If it is sold for 20% discount, then the selling price of the book is:

(a) Rs 240 (b) Rs 260 (c) Rs 230 (d) Rs 280

(vi) If the rate of conversion of 1 Saudi Riyal = Rs 27.15,

then Rs 54,300 = ______ Saudi Riyals.

(a) 1900 (b) 2,100 (c) 2,000 (d) 2,010

(vii) The tax imposed on income is called:

(a) Ushr (b) Jazya (c) Income Tax (d) Property Tax

(viii) Property of a deceased person is to be distributed among a widow, three daughters and two sons. Each son will get a share:

(a) $\frac{1}{8}$ th (b) $\frac{1}{4}$ th (c) $\frac{1}{6}$ th (d) $\frac{1}{3}$ rd

(ix) A sum of money would triple itself at 8% rate of markup in_______years:

(a) 40 (b) 30 (c) 20 (d) 25

(x) A single discount equivalent to successive discounts of 10% and 5% is:

(a) 15% (b) 14.5% (c) 15.5% (d) 14%

2. A company earns a profit which is to be divided among four partners A, B, C and D in the ratio 3 : 4 : 5 : 6. If D gets Rs 54,00, how much will A and B get? What is the total profit?

3. If 30persons use 20 kg of sugar in 50 days, find in how many days 15 persons will use 320 kg of sugar?

4. Which is the greater discount, a single discount of 25% or two successive discounts of 15% and 10% ?

5. Akber purchased a car in Rs 1,000,000 and sold it to Asgharon 20% profit. Then Asghar sold it to Anwar on 20% loss. Find the price paid by Anwar?

6. Usama paid an amount of Rs 40,500 as the first premium of one year for insurances of his car. How much is the price of his car, if the rate is 3.85% and Rs 2,000 are service charges?

7. A factory marked prices of the confectionary items at 20% above the cost price and SAD them to the confirm at some discount. The cost price of confectionary items is Rs 2,500 and the selling price is Rs 2,700. Find the discount % given to the customer.

8. Find the net annual income of Ahmad if he pays Rs 8,900 as income tax and Rs 20,000 as Zakat.

SUMMARY

- Compound proportion is the relationship between three or more proportions.
- When two or more persons run a business together it is called partnership.
- When a person dies then the assets left by him/her is called inheritance to be distributed among his/her legal heirs.
- A bank which accepts deposits, provides loans and services to the clients is known as commercial bank.
- An account on the basis of profit and loss sharing is known as a PLS account.
- Current deposit account is a running account without any interest. It is used by businessmen for managing daily transactions.
- A cheque is a written order that instructs a bank to pay the specific amount from a specified account to the holder of the cheque.
- A Demand Draft is a written order drawn by one bank to another bank, to pay a specified sum of money to the holder of DD on demand.
- Pay order is a document which instructs a bank to pay a certain amount to a third party. The bank which issues a pay order provides a guarantee that the payment will be made.
- On-line banking is the use of instructions by the banks to assist their customers to withdraw money, make deposits, transfer money, check balances, etc.
- An Automated Teller Machine (ATM) is an electric device that allows bank customers to draw cash, check their balances

(at any time) and transfer amounts from one account to other.

- Credit card is a plastic card with a magnetic strip which stores information related to the account holder in a particular bank. These are used for purchasing without cash up to certain limit.
- Markup is the extra money which a bank receives from a client on borrowed money.
- The rate at which the bank gives loan to its account holder is called mark-up rate.
- Over Draft (OD) is a facility provided by a bank to the account holder to withdraw some amount in excess of their original balance (only once).
- **I = PRT,** where P = Principal amount, R = Rate of markup
  T = Time period and I = Markup
- A lease is an agreement between the leasee (user) to pay some amount to leaser (owner) for the use of an asset.
- Discount means reduction in the marked price of an article
  Discount = Marked Price − Sale Price.
- Chain discount means giving discount again and again at various rates.
- Life insurance is an agreement between the insurance company and the policy owner for an agreed period of time and amount.
- Income tax is imposed on the amount of income of a person or firm when the income exceeds a certain limit which is determined by the government every year.
- Exempted income is the income which is not subject to income tax, (such as Agricultural income, pension, property owned by widows).
- Taxable Income = Annual Income − Exempted Income.
- Rebate is the non-taxable income.
- Assessee is a person by whom any tax is payable under the income tax law.